Correlation Under Non-Linearity

2026-10-21

Simulation:

```
simCorr <- function(n){
  dat.null <- mvrnorm(n, c(0,0), matrix(c(1, 0, 0, 1), nrow = 2, byrow = TRUE))
  dat.null.sqr <- dat.null; dat.null.sqr[,2] <- dat.null.sqr[,2]^2

  dat.corr <- mvrnorm(n, c(0,0), matrix(c(1, 1, 1, 1), nrow = 2, byrow = TRUE))
  dat.corr.sqr <- dat.corr; dat.corr.sqr[,2] <- dat.corr.sqr[,2]^2

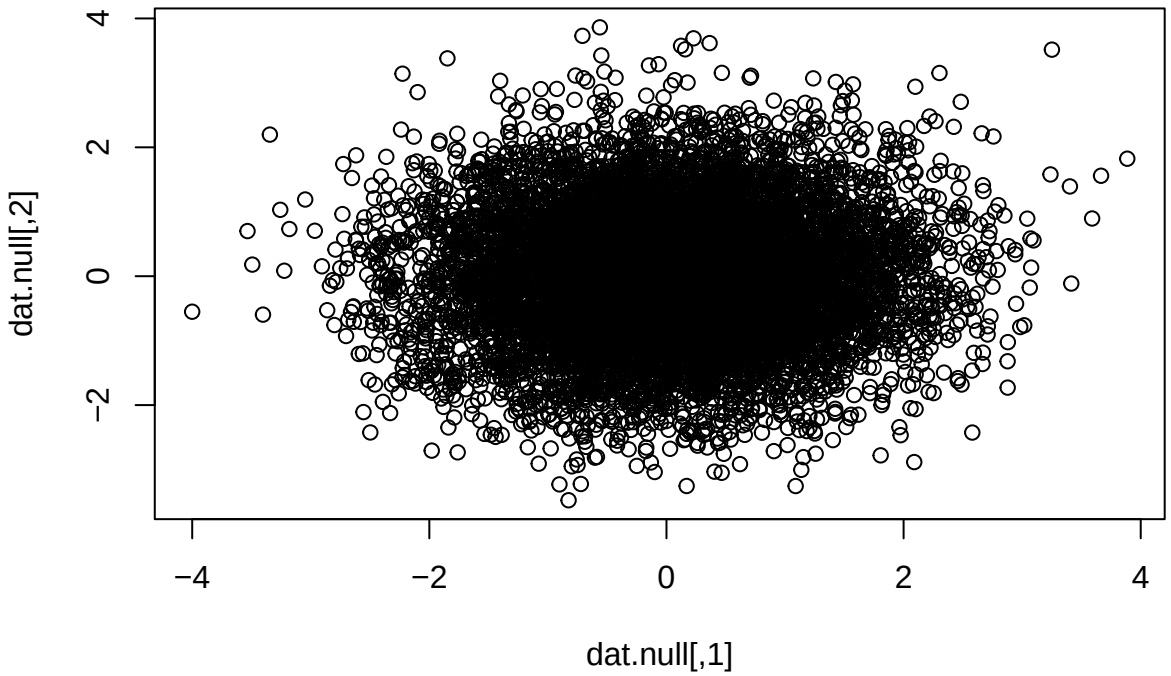
  pval.null <- cor.test(dat.null[,1], dat.null[,2],
                        method = "pearson")$p.value
  pval.null.nonlin <- cor.test(dat.null.sqr[,1], dat.null.sqr[,2],
                               method = "pearson")$p.value
  pval.corr <- cor.test(dat.corr[,1], dat.corr[,2],
                        method = "pearson")$p.value
  pval.corr.sqr <- cor.test(dat.corr.sqr[,1], dat.corr.sqr[,2],
                            method = "pearson")$p.value

  pvals <- c("null" = pval.null, "null.nonlin" = pval.null.nonlin,
             "corr" = pval.corr, "corr.sqr" = pval.corr.sqr)
  return(pvals)
}

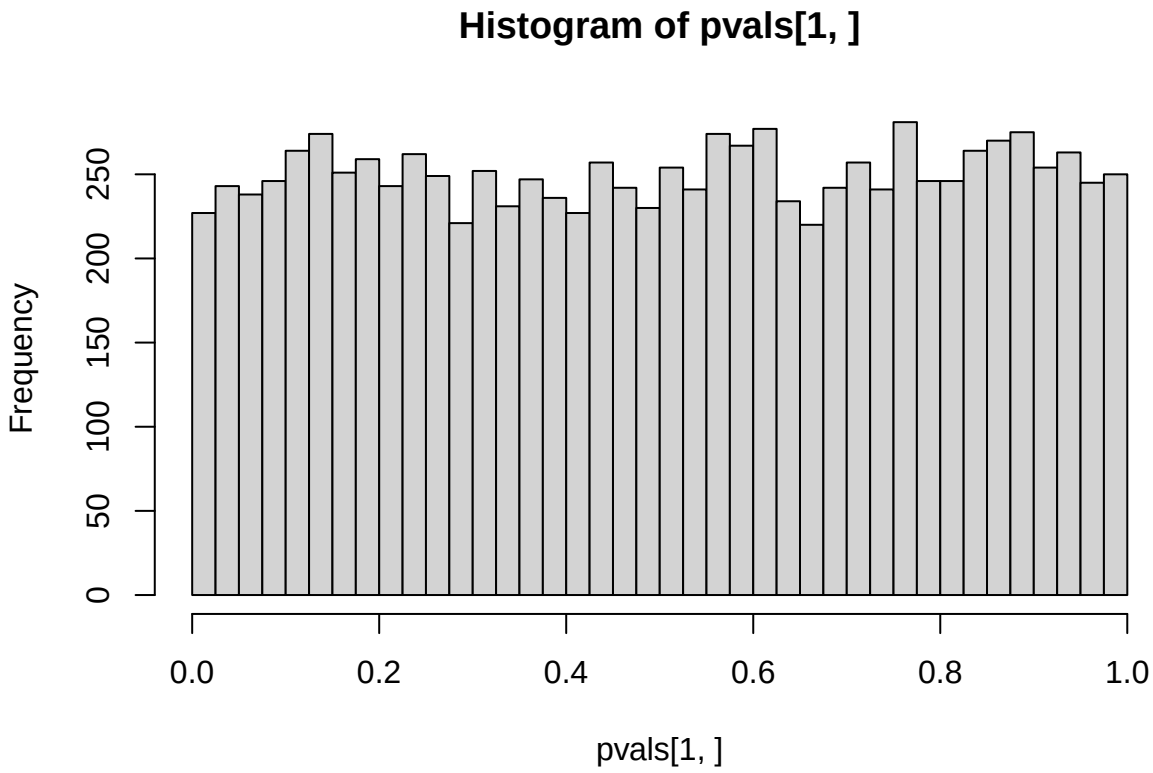
n <- 10000
pvals <- replicate(n, simCorr(2500))
```

Case 1: Centered Null Distribution

```
dat.null <- mvrnorm(n, c(0,0), matrix(c(1, 0, 0, 1), nrow = 2, byrow = TRUE))
plot(dat.null)
```

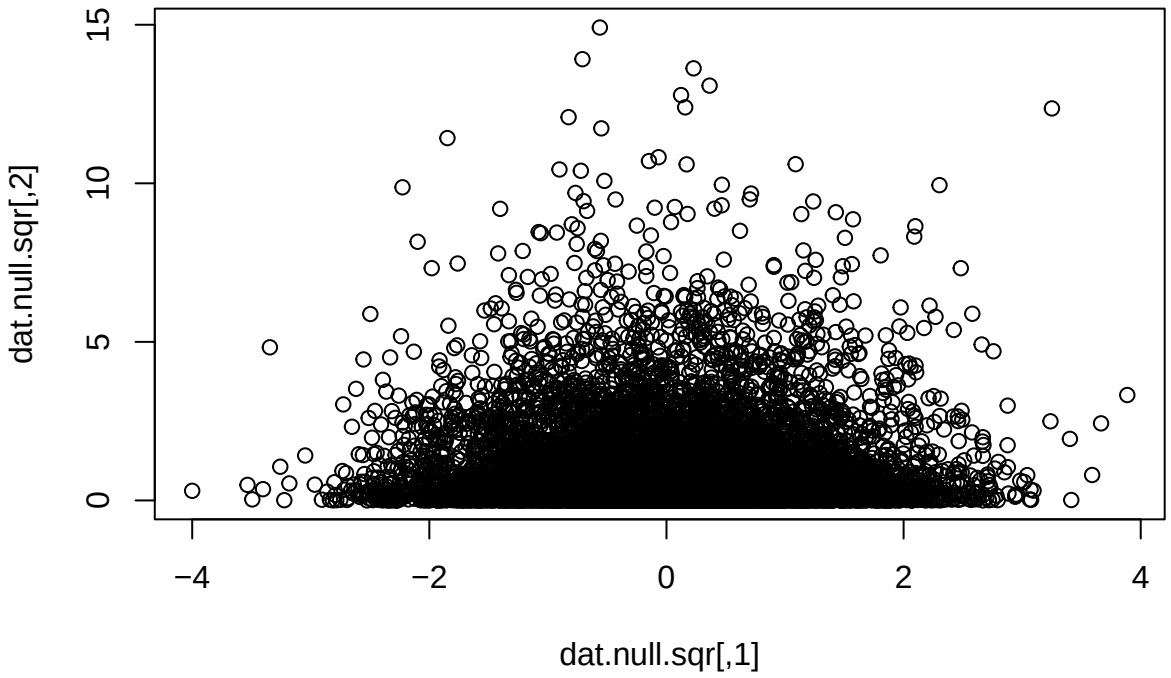



```
hist(pvals[1,], breaks = seq(0,1,length.out = 41))
```

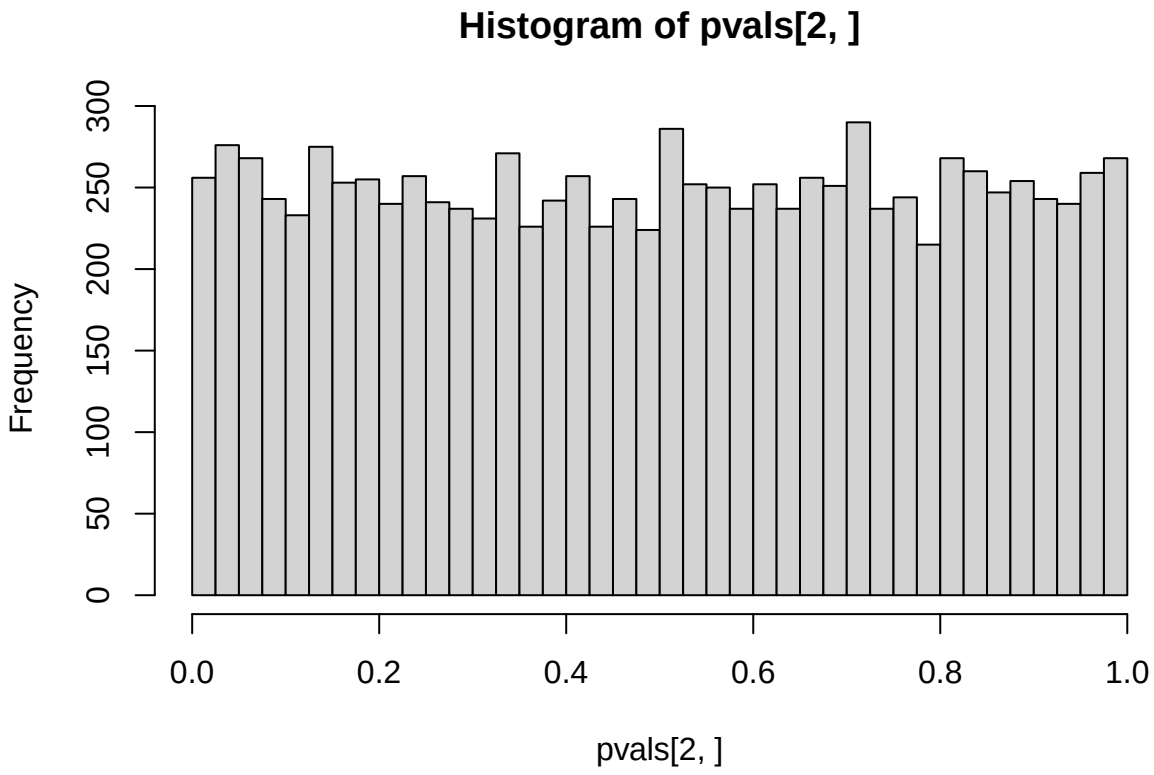


Case 2: Squaring the Null Distribution (Non-linearity in one marginal)

```
dat.null.sqr <- dat.null; dat.null.sqr[,2] <- dat.null.sqr[,2]^2
plot(dat.null.sqr)
```

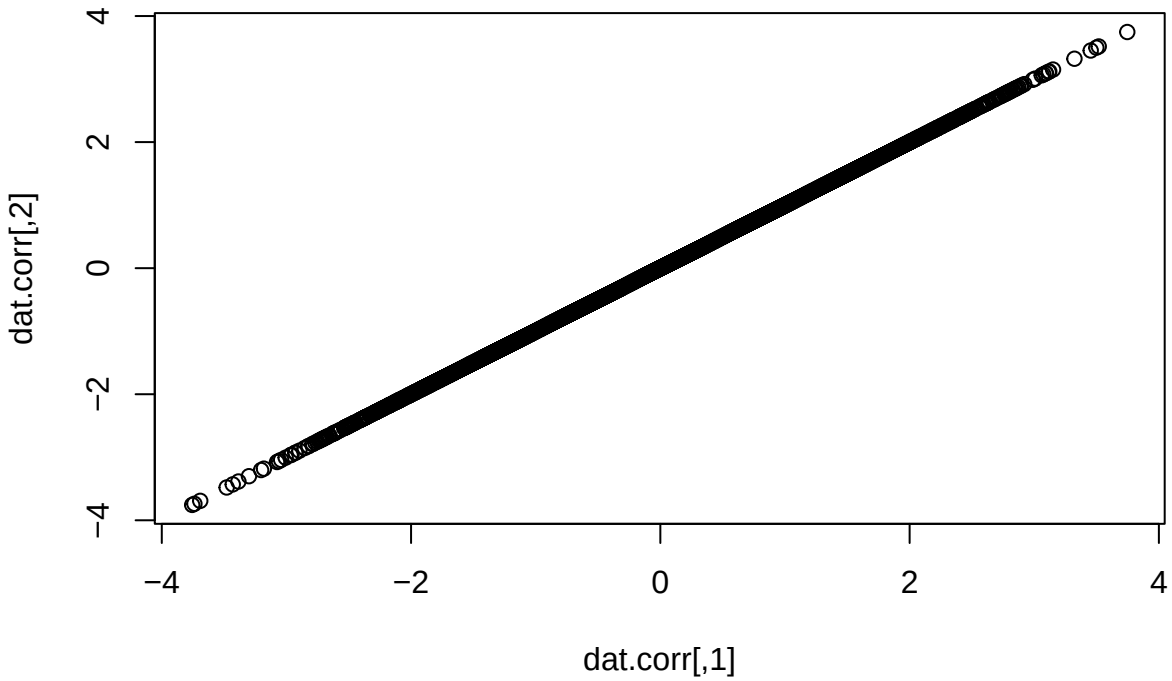



```
hist(pvals[2,], breaks = seq(0,1,length.out = 41))
```

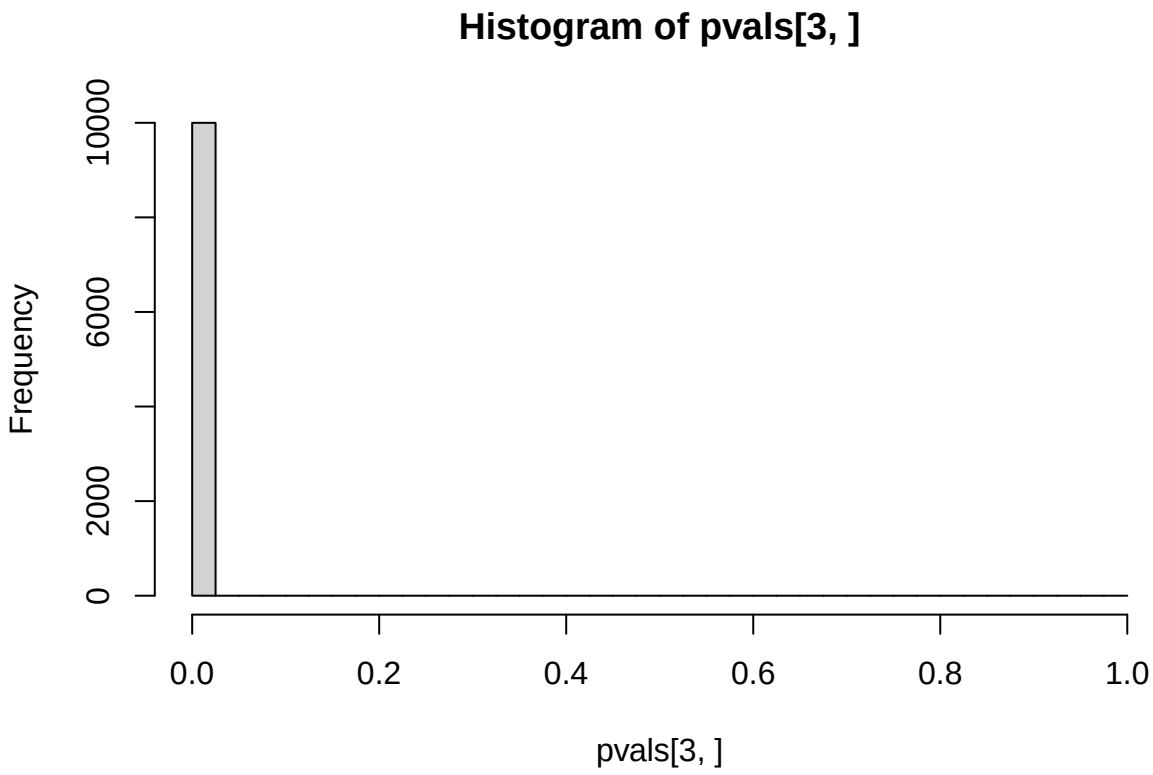


Case 3: Correlated Data

```
dat.corr <- mvrnorm(n, c(0,0), matrix(c(1, 1, 1, 1), nrow = 2, byrow = TRUE))
plot(dat.corr)
```

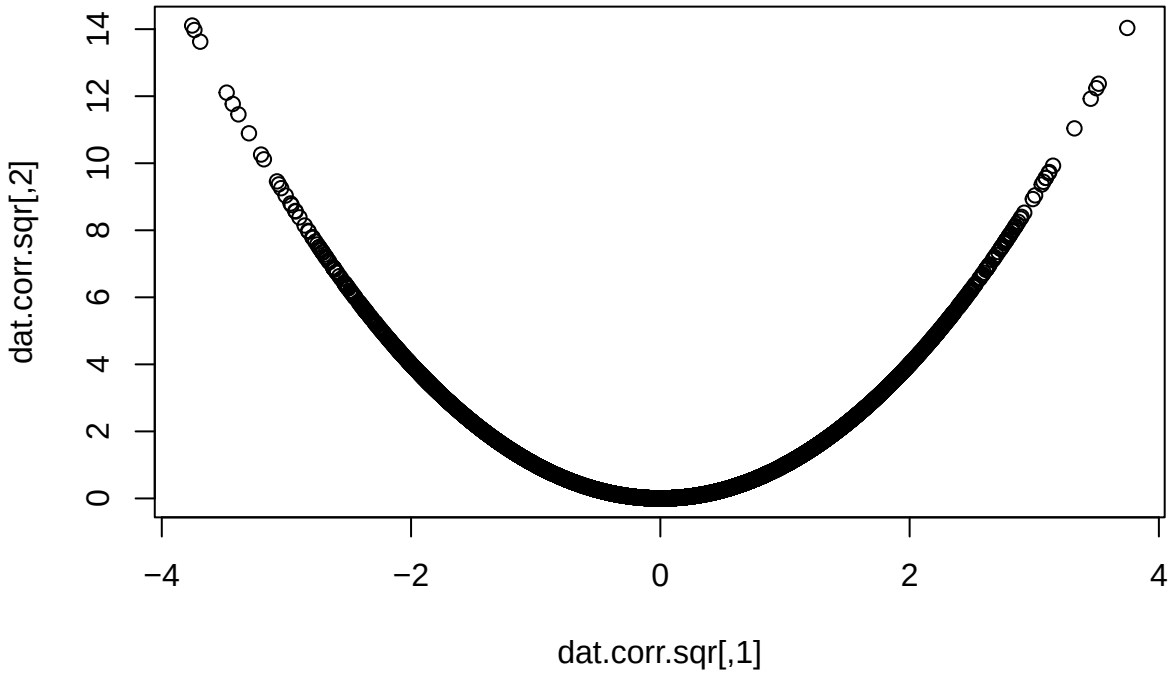



```
hist(pvals[3,], breaks = seq(0,1,length.out = 41))
```



Case 4: Squaring the Correlated Data(Non-linearity in one marginal)

```
dat.corr.sqr <- dat.corr; dat.corr.sqr[,2] <- dat.corr.sqr[,2]^2
plot(dat.corr.sqr)
```




```
hist(pvals[4,], breaks = seq(0,1,length.out = 41))
```

